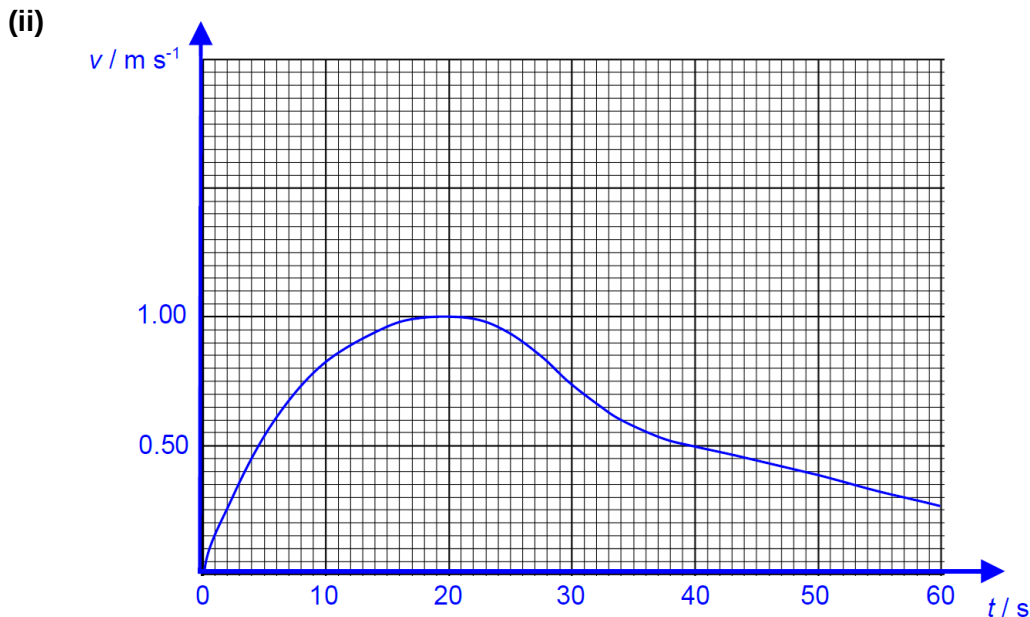


2 (a) (i) An object falling at terminal velocity when the downwards, **B1**
gravitational force is equal in magnitude with the upwards resistive force. **B1**

(ii) An object at its highest point when thrown vertically upwards on Earth or an **B1**
 object at the position of maximum displacement in a simple harmonic motion
 or any reasonable example. **B1**

(iii) Object undergoing uniform circular motion, **B1**
acceleration is perpendicular to velocity therefore speed remains the same **B1**
 but direction of velocity changes.

(b) (i) Maximum speed occur at 20.0 s. **M1**
 Maximum speed = Area under graph from 0.0 – 20.0 s
 $= \frac{1}{2} (0.100)(20.0)$
 $= 1.00 \text{ m s}^{-1}$ **A1**



- Correct shape of graph ($t = 0.0$ to $t = 20.0 \text{ s}$) – **B1**
- Correct shape of graph ($t = 20.0$ to $t = 30.0 \text{ s}$) – **B1**
- Correct shape of graph ($t = 30.0$ to $t = 40.0 \text{ s}$) – **B1**